

Bi-Level Multi-Objective Optimization of Land and Water Resources in the Water-Energy-Food-Carbon Nexus: Application to the Minjiang River Basin

Mingkang Yuan^a, Yangxi Lv^a, Gavin Kader^{b *}, Peng Lei^b, Xiaobing Qu^a

a: College of Management Science, Chengdu University of Technology, Chengdu, 610059, China.

b: Southwestern University of Economics and Finance, Chengdu, 611130, China.

*: Corresponding author: Gavin Kader

Email: gav.kader@outlook.com

Southwestern University of Finance and Economics, No.555 Liutai Avenue, Wenjiang District, Chengdu, 611130, China

Abstract

This study addresses the critical challenge of optimizing agricultural resource allocation within the Water-Energy-Food-Carbon (WEFC) nexus framework. The study employs an innovative bi-level multi-objective programming model to analyze the complex trade-offs between basin-level management and lower-level regional development in the Minjiang River Basin, China (MRB). The model integrates economic, environmental, and social objectives through fuzzy goal programming and elite strategy genetic algorithms. Results demonstrate robust system performance across different weights between upper- and lower-level decision-makers, achieving stable economic benefits ($287.706 \times 10^8 \sim 289.250 \times 10^8$ RMB), sustained food production efficiency ($1.655 \sim 1.694$ kg/m³), and controlled carbon emissions intensity ($5.649 \sim 5.758$ kg CO₂e per kg of agricultural production), reflecting the efficiency of agriculture on the environment. Our analysis reveals distinct spatial patterns in resource allocation and environmental impact, with economically developed regions showing higher resource utilization and emissions compared to the more upstream Aba Prefecture. The optimal

weight (0.5,0.5) between decision levels achieves balanced outcomes, maintaining food security ($989.821 \times 10^4 \text{kg}$ production) while ensuring equitable resource distribution (Gini coefficient: 0.200). The study identifies significant regional variations in carbon management potential, from high emissions in developed areas to substantial sequestration capabilities (27.72%) in mountainous regions. We propose targeted policy recommendations for optimizing agricultural resources in the MRB, including establishing differentiated resource allocation mechanisms that implement zone-specific strategies, such as optimizing intensive agriculture in high-resource zones (e.g. Chengdu Plain) and prioritizing ecological protection in low-resource zones (e.g. Aba Prefecture). Additionally, we recommend developing collaborative carbon management systems with region-specific emission reduction targets whilst enhancing coordination among basin-level and regional authorities. Overall, these measures aim to address regional disparities, improve carbon emission control, and promote sustainable agricultural development.

Keywords: Water-energy-food-carbon nexus; Bi-level optimization; Water-land allocation; Agricultural sustainability; Carbon emissions

1. Introduction

Agriculture, as the foundation of human society's development, plays a crucial role in achieving the United Nations Sustainable Development Goals (SDGs). However, by 2030, continued global population growth is projected to increase food demand by 35-50% [1], energy demand by 20-30%, and water demand by 40% [3]. Meeting these demands poses severe challenges to agricultural sustainability, placing corresponding SDGs – Zero Hunger (SDG2), Clean Water and Sanitation (SDG6), and Affordable and Clean Energy (SDG7) – at the forefront of the United Nations' sustainable development agenda. Agricultural activities, primarily irrigation, fertilization, and cultivation, involve the consumption of water, land, and energy resources, leading to substantial carbon emissions. Large-scale crop production contributes to 34% of global greenhouse gas emissions (GHGs), a proportion increasing by 1% annually, potentially undermining global carbon neutrality objectives [4,5]. Consequently, developing low-carbon agriculture has become a critical pathway for mitigating GHGs while ensuring basic food security [6,7].

The Water-Energy-Food (WEF) nexus represents an integrated approach to resource management that explicitly recognizes the complex linkages between water resources, energy consumption, and food production systems [4,5,7]. In agricultural contexts, this nexus manifests itself through irrigation water requirements, energy inputs for cultivation, and resulting food outputs, all connected through complex feedback mechanisms. The nexus approach fundamentally shifts analytical focus from sectoral optimization to system-wide efficiency, recognizing that isolated improvements in one sector may create unintended consequences in others. Research on WEF nexus has evolved along three distinct trajectories. First, conceptual assessment frameworks have been developed, exemplified by Feng et al. [8], which conducted comprehensive analysis of cross-sectoral pressure transmission mechanisms in Central Asia, and Li et al. [9], which investigated vulnerability factors in China's agricultural systems. Second, optimization approaches have been advanced by researchers like Zhang et al. [10] and Wang et al. [11], who balanced water availability against food production targets while minimizing energy inputs in arid and semi-arid regions. Third, the integration of climate and carbon considerations has expanded the traditional framework, with Chen et al. [14] incorporating climate change impacts and evaluating adaptation strategies under varying scenarios. These studies collectively demonstrate that optimizing nexus relationships through adjusted irrigation strategies and modified planting structures can significantly improve resource efficiency while ensuring food security and reducing environmental impacts - a holistic approach that surpasses traditional sectoral boundaries. This perspective forms the foundation for the Water-Energy-Food-Carbon (WEFC) nexus framework adopted in this study.

Given the intensification of global climate change and the urgency of sustainable development goals, incorporating carbon emissions into the **traditional** WEF nexus has emerged as a critical **advancement** in sustainability research [15,16,17]. This integration acknowledges the intricate connections between agricultural systems, resource utilization, and climate impacts [18]. The World Economic Forum identifies CO₂ as the primary greenhouse gas contributor, with increasing atmospheric concentrations directly affecting crop growth **conditions and ecological stability** [19]. Integrating carbon emissions as a fourth dimension, augmenting to the WEFC nexus, allows for a more holistic evaluation of agricultural trade-offs which can be combined with the design of management strategies that are aligned with global climate objectives vis-à-vis the SDGs [6,13].

Recent studies have made significant progress in operationalizing the WEFC nexus framework. Li et al. [20] pioneered a water-energy-food-ecosystem model that

explicitly incorporates carbon emissions to analyze how different resource allocation schemes affect carbon footprints. Zhang et al. [21] investigated the synergistic relationships among water utilization, energy consumption, and carbon emissions in agricultural production systems, providing valuable insights for low-carbon agricultural development pathways. Feng et al. [22] demonstrated practical applications through an agricultural sustainability optimization model based on WEF nexus and CO₂ emissions in the Tarim River Basin. These studies collectively illustrate how integrating agricultural water, energy, food, and carbon emissions into a unified analytical framework enables more nuanced assessment of agricultural activities' impacts [23], facilitating the development of targeted sustainable management strategies [24,25].

The complexity and systemic nature of the agricultural WEFC nexus necessitates advanced research methodologies to elucidate its underlying mechanisms [26,27]. Issues of agricultural resource scarcity, land degradation, and water shortages continue to intensify [28], with these interrelated challenges severely affecting food production capacity [29]. Effective allocation of limited resources has consequently become a central focus in agricultural system management [30,31], particularly when balancing the interests of multiple stakeholders and addressing environmental, social, and economic concerns [32]. Computational modeling provides a scientific foundation for exploring the inseparable relationships among nexus elements [33], while mathematical programming enables quantitative analysis of correlations [14], risks [34], and feedback mechanisms [35]. Multi-objective optimization approaches have proven especially valuable, allowing researchers to simultaneously consider multiple competing objectives [36], achieve optimal balance among conflicting goals [37], and optimize resource allocation and cropping structures [5,17], particularly within complex agricultural irrigation systems [13,22].

While valuable contributions have emerged in water-land resource allocation and carbon emission management within agricultural systems [18,21], **significant gaps remain in applying these methods to multi-level decision-making frameworks [16,38]. Current literature often emphasizes basin managers as primary macro-level decision-makers, with somewhat less attention devoted to regional governments as regional-level actors [39,40]. This highlights a critical research gap: the need for models that explicitly account for hierarchical levels, their respective priorities, and their strategic interactions when seeking sustainable outcomes across economic, social, and ecological dimensions.**

Bi-level programming offers a framework for addressing interest differences among hierarchically distinct stakeholders and exploring balanced approaches in

complex systems [41,42]. Its approach allows for consideration of multiple objectives across different decision-making levels while working toward equilibrium solutions [41]. Despite progress in WEFC nexus research, key areas require further attention to enhance (theoretically, and practically) multi-objective optimization models:

- **Hierarchical strategic interactions:** Current studies typically focus on single-level decision optimization (e.g., a central planner), failing to capture the dynamic strategic interactions between different governance levels, such as basin-level authorities (upper-level) and regional-level government (lower-level); this overlooks the conflicting objectives and trade-offs inherent in real-world governance.
- **Inter-regional objective balancing:** Different regions often have different priorities among economic, social, and ecological objectives. Beyond their geographical differences, this means agricultural systems exhibit significant spatial heterogeneity. By fully incorporating heterogeneity into a model allows for a more equitable and efficient allocation of resources across different regions within a larger area.
- **Uncertainty considerations:** Despite existing studies incorporating probability distributions and fuzzy sets to model particular uncertain parameters, there remain opportunities to model decision-making that incorporates this uncertainty in the context of modeling the impacts of variability in terms of water availability fluctuations as a result of climate change.

Building upon existing research, this study seeks to develop a bi-level multi-objective nonlinear programming model that considers agricultural water-energy-food nexus dynamics alongside carbon emission management. The model aims to account for the different priorities that might exist between basin managers and regional governments within China's western Minjiang River Basin. The approach suggests positioning basin managers as upper-level decision-makers concerned with system economic efficiency, water resource utilization, and greenhouse gas emissions of the overall basin, while regional governments may function as lower-level decision-makers focusing on farmer income distribution, food production targets, and agricultural benefits within their jurisdictions. In this study, we employ a Stackelberg game framework (a hierarchical decision-making structure from game theory) to model the strategic interactions between governance levels. Under this framework, upper-level basin managers act as 'leaders', establishing basin-wide policies and resource allocation strategies, while lower-level regional governments serve as 'followers' who

optimize their operational decisions in response to the upper-level decisions. This hierarchal structure acknowledges the potential interdependence between decision-making levels, with their interactions potentially contributing to sustainability goals including carbon emission management, agricultural income, and food security.

The solution to this bi-level multi-objective nonlinear optimization problem falls within the category of Non-deterministic Polynomial-time hard (NP-hard) problems [16,42], which typically involve simultaneous consideration of objectives at multiple hierarchical levels [16,23]. Traditional algorithms may face certain challenges when addressing problems of this nature [23]. To address these considerations, we explore the application of bio-inspired algorithms that have shown promise in handling complex optimization problems [42,43]. Specifically, we propose an approach that draws on elements of the Non-dominated Sorting Genetic Algorithm framework enhanced with fuzzy goal programming concepts. This approach utilizes membership functions to represent objectives at both decision levels [41,43,44], attempting to balance broader optimization priorities with local considerations while seeking to enhance resource management and system sustainability [16,45,46].

Table 1 presents a comparative analysis of existing research, demonstrating the comprehensiveness of this study's approach. Overall, the main contributions of this study lie in constructing a scientific and reliable decision support system aimed at assisting agricultural decision-makers in formulating rational strategies within the complex Water-Energy-Food-Carbon (WEFC) nexus management environment. To reiterate, the reviewed literature reveals distinct interpretations and emphases within the WEF nexus framework across different research contexts. Studies like Feng et al. [22] and Li et al. [20] emphasize conceptual assessment approaches, focusing on cross-sectoral pressure transmission mechanisms and vulnerability factors. In contrast, optimization-focused research exemplified by Zhang et al. [21] and Wang et al. [11] prioritizes resource efficiency maximization through balancing water availability against food production targets while minimizing energy inputs in water-scarce regions. Climate-integrated studies such as Chen et al. [14] expand the traditional WEF framework by incorporating climate change impacts and adaptation strategies under different environmental scenarios. Other research has focused on improving and diversifying the use of the WEFC nexus. Li et al. [20] developed a Water-Energy-Food-Ecosystem model that explicitly incorporates carbon emissions to analyze resource allocation impacts on carbon footprints, while Zhang et al. [21] investigated synergistic relationships among water utilization, energy consumption, and carbon emissions to identify low-carbon agricultural development pathways. Our study uniquely

contributes to this by integrating a bi-level decision-making framework that models strategic interactions between basin managers and regional governments within the WEFC nexus, addressing the gap in multi-level optimization approaches that previous studies have not systematically explored.” Thus, our specific innovations include:

(1) Integration of carbon emissions into the agricultural Water-Energy-Food nexus through a bi-level programming framework, creating a comprehensive WEFC-nexus optimization model that addresses both upper-level (basin level) and lower-level (regions within the basin) decision-making dimensions, offering new analytical pathways for agricultural sustainability assessment; (2) Development of a structured approach for modeling the strategic interactions between basin managers and regional governments, enabling collaborative optimization of competing objectives across multiple decision levels through specialized membership functions and interactive decision mechanisms; (3) Building on the NSGA-II algorithm, we introduce a novel optimization framework that integrates fuzzy programming to model the bi-level (Stackelberg) interactions between upper-level basin managers and lower-level regional governments. The innovation of our approach lies in coupling fuzzy membership functions to model the hierarchical structure of the stakeholders, which directly captures the strategic interactions between decision-making levels.

Table 1 Summary of key literature

Key literatures	Bi-level	WEF-Nexus	Carbon emissions	Spatial difference	Uncertainty	Gini	Heuristic algorithm	Fuzzy planning
Gómez-gardars et al.[47]		√	√	√			√	
Ogbolumani et al.[48]; Ogbolumani et al.[49]		√	√		√		√	
Feng et al. [22]		√	√	√	√		√	
Zheng et al. [34]	√	√		√	√	√		√
Li et al. [20]		√	√	√		√		
Fitton et al. [50]	√	√	√		√		√	
Tu et al. [41]	√			√	√		√	√
Meng et al. [51]		√		√		√		√
Ji et al.[40]		√	√	√	√		√	
Xu et al. [52]	√	√			√	√	√	
This study	√	√	√	√	√	√	√	√

The remainder of this study is organized as follows: Part 2 focuses on the study area and data collection methods; Part 3 covers the methodology, including the analytical framework, model building and solution algorithms. Part 4 analyzes the results and discussion; Part 5 summarizes key findings and conclusions.

2. Materials

2.1 Study area

This study focuses on the Minjiang River Basin (MRB), located in the upper reaches of the Yangtze River in southwestern China (Fig. 1). The Minjiang River, the largest tributary of the Yangtze River in terms of water volume, originates from the southern foothills of the Min Mountains, with a total length of 1,279 kilometers and a basin area of 133,500 square kilometers. The basin is characterized by abundant water resources, humid climate, and subtropical conditions, with an average annual runoff exceeding 90 billion cubic meters and hydropower resources accounting for one-fifth of the Yangtze River system. Based on geographical characteristics along the main stream, the basin is divided into three sections: the upper reaches, located above Dujiangyan City; the middle reaches, extending from Dujiangyan City to Leshan; and the lower reaches, from Leshan to Yibin. The river flows through Aba Tibetan and Qiang Autonomous Prefecture, Chengdu, Meishan, Ziyang, Ya'an, Leshan, and Zigong cities before finally joining the Yangtze River in Yibin City. With continuous socio-economic development and population growth, water resource demand has increased annually. Simultaneously, agricultural water consumption has risen significantly due to expanding irrigation areas and environmental pressures [53], leading to water scarcity issues in the MRB, particularly during the dry seasons, which threatens both agricultural output and ecological water requirements [54].

Within the MRB, the water resource utilization rate has reached 48.5%, exceeding the internationally recognized 40% threshold for safe water resource development. Furthermore, inappropriate farming practices have reduced the basin's water storage capacity, resulting in severe land erosion problems [55]. This highlights the imbalance between water and land resources in the MRB. Therefore, it becomes essential to (i) thoroughly investigate the intrinsic connections among water, energy, and food resources in the MRB, (ii) comprehensively evaluate the optimal allocation of various elements, and (iii) explore the evolution patterns of the WEFC nexus system.

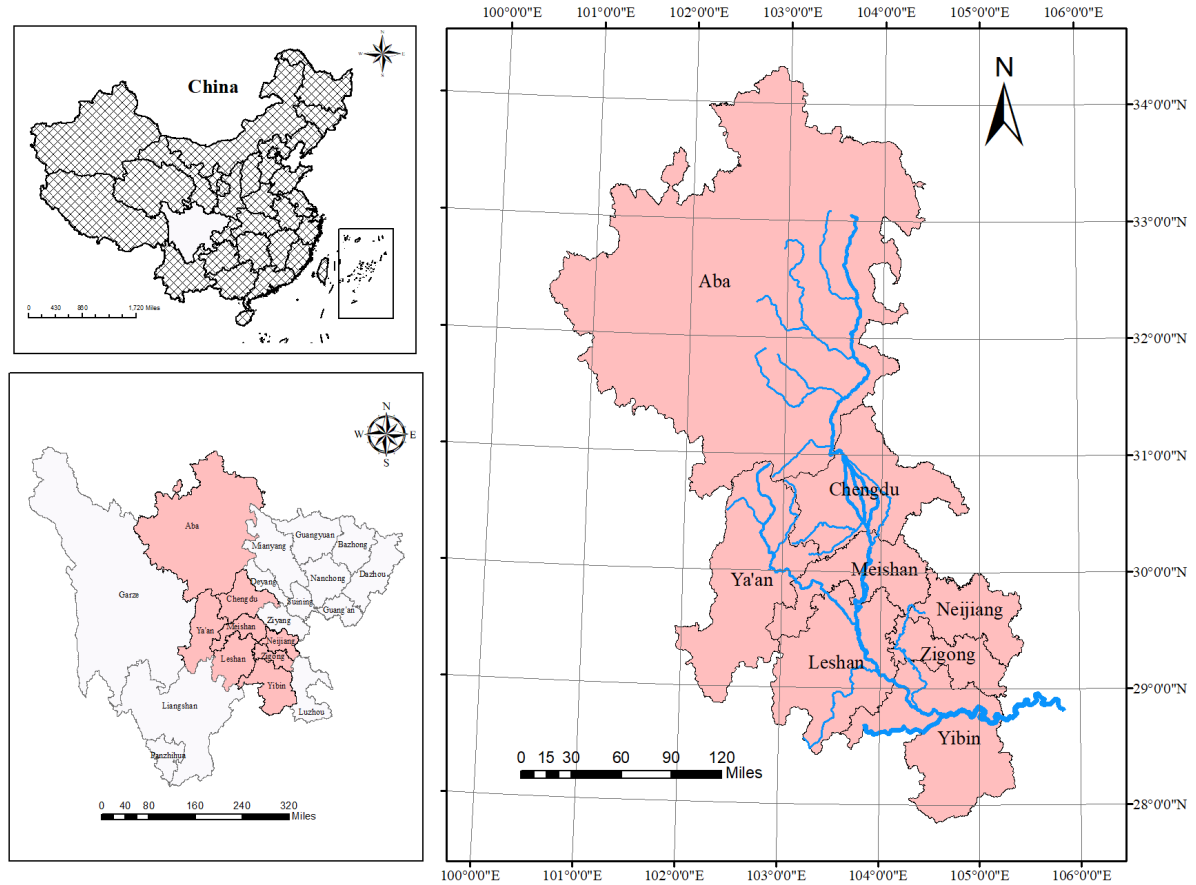


Fig.1. Map of the MRB

2.2 Data collection and processing

The WEFC nexus in agricultural systems represent a complex system involving interactions among numerous parameters. To thoroughly investigate these relationships, this study collected research data on Sichuan province related to water, energy, and food resources from relevant statistical websites, statistical yearbooks, government reports, and related research papers, based on model definitions and associated variables.

Water resource data (Table. 2) includes: agricultural water consumption; irrigation water usage per unit area; ratio of irrigated area to total planted area. These data were sourced from regional water resource bulletins. Food resource data (Tables 3 & 4) includes: grain yield per unit area; per capita food demand; crop planting area. These data were primarily obtained from the Sichuan Province Statistical Yearbook. Energy resource data (Table. 5) includes: diesel consumption per unit area; agricultural electricity consumption per unit area. These data were sourced from the Sichuan Province Statistical Yearbook. In addition, other data on water resources and data related to carbon emissions from agriculture are placed in the supplementary material (Appendix C) file of this study.

Table 2 The agricultural water consumption of different parts of the subarea in MRB

Subarea	X_i	ξ_i^{loss}	Agricultural water consumption max	Agricultural water consumption min	ε_{ij}
Chengdu	255.3532	0.18	329123	241400	0.508
Zigong	70.9293	0.21	44226	22949	0.272
Neijiang	63.4354	0.19	62441.25	31847.03	0.278
Leshan	160.2989	0.29	101389	61652	0.405
Meishan	203.4407	0.31	76020	23524	0.534
Yibin	82.4405	0.19	86054	47372	0.314
Ya'an	207.2384	0.26	49848.67	3260.3	0.459
Aba	12.1871	0.24	15378	13016	0.393

Table 3 Data on crop production and food consumption in the MRB

Subarea	$Y_{i1}(10^3\text{kg/ha})$	$Y_{i2}(10^3\text{kg/ha})$	$Y_{i3}(10^3\text{kg/ha})$	$Y_{i4}(10^3\text{kg/ha})$	$FD_i(10^3\text{kg})$
Chengdu	8.02	4.766	5.137	2.324	192.410
Zigong	8.71	3.322	5.373	2.743	644.030
Neijiang	7.303	3.212	5.489	2.635	424.551
Leshan	7.239	3.022	4.131	1.975	354.986
Meishan	7.787	4.117	5.314	2.215	368.260
Yibin	8.097	2.838	5.285	2.645	771.760
Ya'an	7.127	3.594	5.154	1.621	237.012
Aba	4.857	3.359	4.213	1.943	178.360

Table 4 Maximum area under crop j in subarea i in the MRB

Subarea	$Tk_{i1}(10^3\text{km}^2)$	$Tk_{i2}(10^3\text{km}^2)$	$Tk_{i3}(10^3\text{km}^2)$	$Tk_{i4}(10^3\text{km}^2)$
Chengdu	233.24	106.03	44.77	41.46
Zigong	81.42	47.70	31.10	52.86
Neijiang	86.05	70.80	59.70	49.95
Leshan	90.11	17.52	49.70	23.28
Meishan	116.66	79.90	40.40	19.40
Yibin	153.48	78.70	70.40	34.72
Ya'an	29.82	15.77	28.93	16.88
Aba	1.00	7.40	15.80	9.69

Table 5 Data related to agricultural electricity in the MRB

Subarea	$q^{elec}(\text{kwh/ha})$	$Q^{des}(\text{t/ha})$
Chengdu	9809.6767	0.4633
Zigong	5861.0093	0.1375
Neijiang	8133.7523	0.1050
Leshan	8774.6254	0.1347
Meishan	4467.5446	0.0808
Yibin	7803.2651	0.1407
Ya'an	9659.1878	0.1356
Aba	9795.3584	0.1319

3. Methods

3.1 Analytical framework

As the challenges of sustainable agricultural development grow increasingly complex, optimizing and managing the WEF-nexus has become a critical issue. The operational efficiency of this system not only directly impacts agricultural production and ecological sustainability but also requires the coordination of interests among multi-level decision-making entities [56]. This study addresses three core challenges: (1) the efficient allocation of water and land resources at the watershed scale, (2) the optimization and regulation of regional crop planting structures, and (3) the synergistic promotion of economic development and environmental protection.

The main challenges faced by current research can be categorized into three aspects. First, there are differences in objectives between watershed management agencies and regional governments in resource allocation decisions, necessitating the establishment of effective coordination mechanisms [41,42]. Second, the imbalance between limited water and land resources and the growing demand for agricultural water use is becoming increasingly prominent [26,29]. Finally, the trade-off between improving agricultural production efficiency and environmental protection, particularly the control of GHGs such as CO₂, requires systematic solutions. Simultaneously, uncertainty is a key issue addressed in this study.

To address these challenges, this study constructs a bi-level multi-objective nonlinear integrated analysis model that effectively reflects the complexity and multi-level decision-making characteristics of agricultural systems and captures the strategic interactions and potential conflicts among decision-makers at different levels. Our model conceptualizes the governance structure as a bi-level hierarchy, positioning watershed management agencies as the upper-level decision-makers (leaders) primarily concerned with the macro-level objectives of the agricultural system, including maximizing overall economic benefits, improving water resource utilization efficiency, and controlling GHGs. At the same time, the model considers regional governments within the watershed as lower-level decision-makers ('followers'), focusing on typical microeconomic objectives such as the equity of farmers' income distribution, ensuring food production, and increasing total agricultural revenue. This approach reflects the complexity and multi-level decision-making characteristics of agricultural systems and captures the strategic interactions among different decision-making entities, providing a scientific basis for achieving sustainable regional agricultural development. The research framework of this study is presented in Fig. 2.

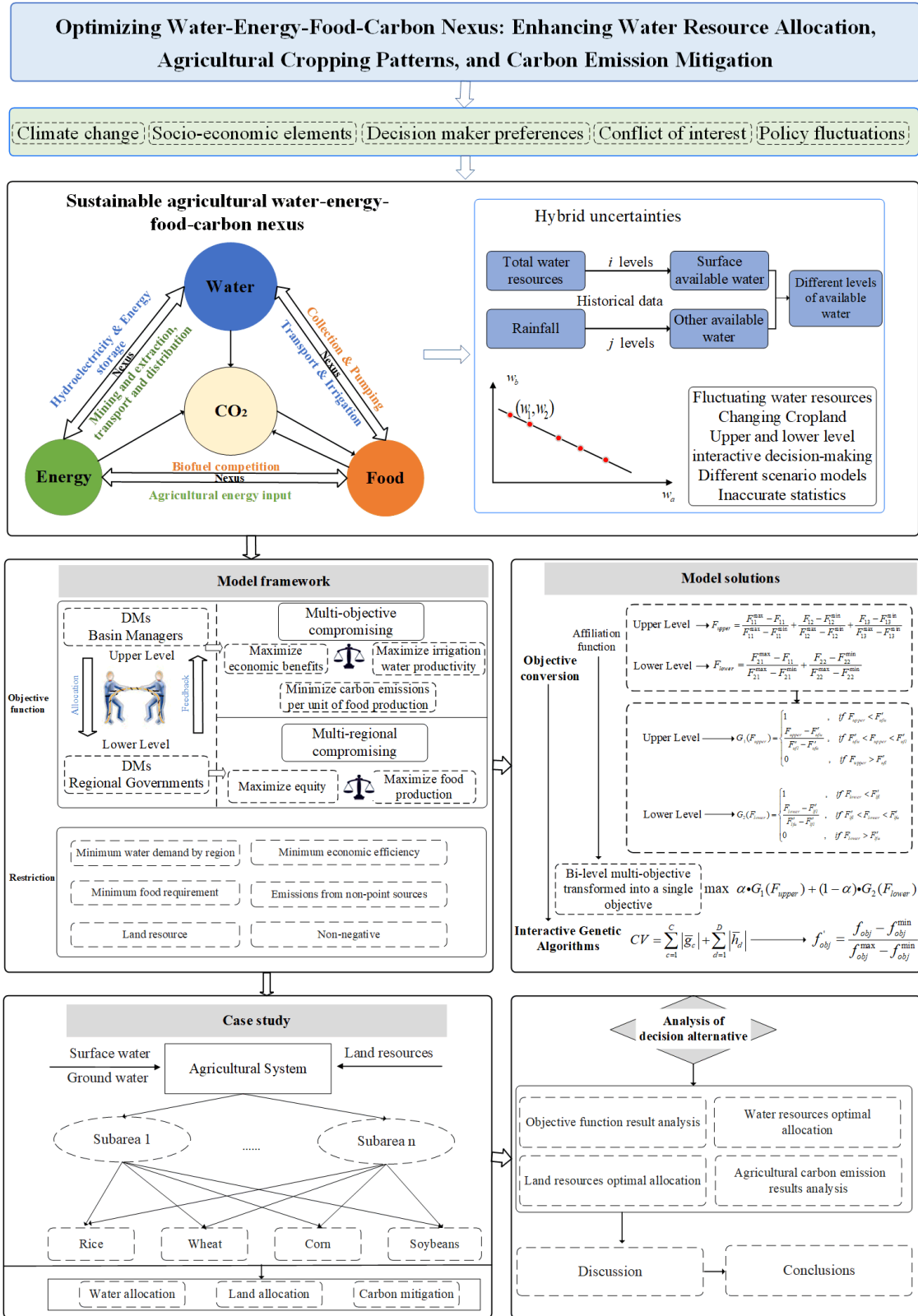


Fig. 2 Analytical framework

3.2 Model Construction

3.2.1 Symbol description

Table 6 contains all the notation used in the model below.

Table 6 Model symbols and variable definitions

Decision variables	Definitions (+ unit of measurement)
DA_i	Water allocation for irrigated agriculture of subarea i (10^8m^3)
A_{ij}	Area planted with crop j in subarea i (10^3km^2)
Carbon footprint symbols	Definitions
ET_j	N_2O emission factor for crop j (kg/ha)
EN_j	N_2O emission factor in crop j nitrogen fertilizer (kg CE/kg)
EF_j	N_2O emission factor in crop j compound fertilizer (kg CE/kg)
q^{fer-n}	Nitrogen fertilizer use per unit area (Ton/ha)
q^{fer-f}	Compound fertilizer use per unit area (Ton/ha)
q^{fer-o}	Potash fertilizer use per unit area (Ton/ha)
q^{fer-m}	Phosphorus fertilizer use per unit area (Ton/ha)
q^{pes}	Pesticide use per unit area (Ton/ha)
q^{film}	Agricultural film use per unit area (Ton/ha)
q^{des}	Fuel use per unit area (Ton/ha)
q^{elec}	Electricity use per unit area for irrigation (KW h/ha)
Cn	Unit price of nitrogen fertilizer (RMB/ton)
Cf	Unit price of compound fertilizer (RMB/ton)
Co	Unit price of potash fertilizer (RMB/ton)
Cm	Unit Price of Phosphorus Fertilizer (RMB/ton)
$Cpes$	Unit Price of Pesticides (RMB/ton)
$Cfilm$	Unit Price of Agricultural Film (RMB/ton)
$Cdes$	Unit Price of Fuel (RMB/ton)
$Celec$	Unit price of electricity for irrigation (RMB/ton)
μ^{fer-n}	Emission factor for nitrogen fertilizer (kg ce/kg)
μ^{fer-f}	Emission factor for compound fertilizer (kg ce/kg)
μ^{fer-o}	Emission factor for potash fertilizer (kg ce/kg)
μ^{fer-m}	Emission factor of phosphate fertilizer (kg ce/kg)
μ^{pes}	Emission factor of pesticide (kg ce/kg)
μ^{film}	Emission factor for agricultural films (kg ce/kg)
μ^{des}	Emission factor for fuel (kg ce/kg)
Carbon footprint symbols	Definitions
μ^{elec}	Emission factor for electricity used for irrigation (kg ce/kg)

ρ^{rice}	Emission coefficient of methane per unit area of farmland (kg/ha)
NM	Proportion of nitrogen content in compound fertilizer (%)
TN	Maximum allowable rate of nitrogen loss (%)
NR_{ij}	Nitrogen demand of crop j in sub-area i
$e\ g$	Empirical coefficients in the carbon sequestration model
β	Proportion of straw to harvested portion (%)
σ	Proportion of return to field (%)
η	Root-crown ratio (%)
v	Percentage of carbon content in roots (%)
Model Parameters	Definitions
i	Subarea i
j	Crop j , $j = 1$: Rice; $j = 2$: Wheat; $j = 3$: Corn; $j = 4$: Soybean
FP_{ij}	Grain production of crop j in watershed subarea i (10^3ton)
ε_{ij}	Proportion of irrigated area to cultivated area in subarea i crop j (%)
χ_{ij}	Irrigation water consumed per unit of irrigated area for crop j in sub-region i (10^3m^3)
Y_{ij}	Yield per unit area of crop j in subarea i (10^3ton)
FD_i	Per capita food demand in subarea i (10^3kg)
POP_i	Population of subarea i (10^4 people)
TK_{ij}	Maximum area under crops j in subarea i (10^3km^2)
ξ_i^{loss}	Rate of water transportation losses in subarea i (%)
MP_j	Economic benefits from crop j per unit of production (MRB)
TNB_i	Net economic benefits of agricultural production in subarea i (MRB)
ESV_i	Value of ecosystem services in subarea i (MRB)
TB_i	Economic benefits from agricultural production in subarea i (MRB)
TC_i	Costs incurred by agricultural production in subarea i (MRB)
EB_i	Average value of economic benefits of agriculture in the last five years in subarea i (MRB)
$E^{Me}[Q(t)]$	Expected availability of water resources in the MRB (10^8m^3)

3.2.2 Objective function

(1) Upper level decision makers optimization

In the bi-level model, the upper-level decision-makers represent watershed management, tasked with optimizing overall performance and environmental sustainability [16, 42]. In this model, the upper-level decision-maker needs to prioritize large-scale impact factors, such as the total economic benefits of the agricultural system, GHGs, and emission reduction potential. By optimizing these objectives, upper-level decision makers can develop more reasonable and effective watershed management strategies to achieve coordinated optimization of the watershed with multiple objectives.

Objective 1: Maximizing economic benefits

The total profit of agricultural production refers to the sum of the economic and ecological values created by agricultural activities over all areas. Agricultural activities not only produce crops to meet human material needs but also provide ecosystem services to maintain the functions of natural resources and the environment [22,24]. Therefore, the total profit of agricultural production includes two aspects. First, the total net economic benefit of crops, which is the total revenue of crops minus the total cost of agricultural production, and second, the value of ecosystem services, which is the economic benefit brought by the protection and improvement of ecosystem elements such as land, water, air, and biodiversity through agricultural activities. The total profit can be expressed by the following formula:

$$\begin{aligned} \text{Max } F_{11} &= \sum_{i=1}^I \text{TNB}_i + \text{ESV}_i \\ \text{TNB}_i &= \text{TB}_i - \text{TC}_i \\ \text{TB}_i &= \sum_{j=1}^J A_{ij} Y_{ij} \text{MP}_j \\ \text{TC}_i &= \sum_{j=1}^J A_{ij} (\text{Cn} \times q^{\text{fer}-n} + \text{Cf} \times q^{\text{fer}-f} + \text{Co} \times q^{\text{fer}-o} + \text{Cm} \times q^{\text{fer}-m} \\ &\quad + \text{Cpes} \times q^{\text{pes}} + \text{Cfilm} \times q^{\text{film}} + \text{Cdes} \times q^{\text{des}} + \text{Celec} \times q^{\text{elec}}) \\ \text{ESV}_i &= f\left(\sum_{j=1}^J A_{ij}\right) \end{aligned} \tag{1}$$

Objective 2: Maximizing irrigation water productivity

Irrigation water productivity, defined as the quantity of agricultural product per unit of irrigation water, reflects the efficiency of water utilization [25,29]. Maximizing irrigation water productivity means producing the most agricultural products with the least water resources, whilst ensuring food security and increasing farmers' income. It can be expressed as follows:

$$\begin{aligned} \text{Max } F_{12} &= \sum_{i=1}^I \sum_{j=1}^J \text{FP}_{ij} / \sum_{i=1}^I (1 - \xi_i^{\text{loss}}) \text{DA}_i \\ \text{FP}_{ij} &= A_{ij} \times Y_{ij} \end{aligned} \tag{2}$$

Objective 3: Minimizing carbon emissions per unit of food production

Carbon emissions from agricultural production processes are an important indicator of environmental impacts, which not only exacerbate global warming, but also affect water resources, ecosystems and socio-economic aspects of watersheds [30,37]. Therefore, controlling and reducing carbon emissions from agricultural production processes is an important and urgent task facing watershed resource managers. The aim of this study is to analyze the sources, quantities and spatial distribution of carbon

emissions from agricultural production processes in the watershed and to minimize carbon emissions from agricultural production processes based on the calculation and analysis of carbon emissions as expressed below:

$$\begin{aligned}
Min F_{13} &= (ECO_2 + 298 \times EN_2O + 25 \times ECH_4 - CS) / \left(\sum_{i=1}^I \sum_{j=1}^J A_{ij} Y_{ij} \right) \\
ECO_2 &= \sum_{i=1}^I \sum_{j=1}^J \left(A_{ij} \cdot \mu^{fer-n} \cdot q^{fer-n} + A_{ij} \cdot \mu^{fer-f} \cdot q^{fer-f} + A_{ij} \cdot \mu^{fer-o} \cdot q^{fer-o} + A_{ij} \cdot \mu^{fer-m} \cdot q^{fer-m} \right. \\
&\quad \left. + A_{ij} \cdot \mu^{pes} \cdot q^{pes} + A_{ij} \cdot \mu^{film} \cdot q^{film} + A_{ij} \cdot \mu^{des} \cdot q^{des} + A_{ij} \cdot \mu^{elec} \cdot q^{elec} \right) \\
EN_2O &= \sum_{i=1}^I \sum_{j=1}^J A_{ij} \cdot (ET_j + q^{fer-n} \cdot EN_j + q^{fer-f} \cdot EF_j) \\
ECH_4 &= \sum_{i=1}^I \sum_{j=1}^J A_{ij} \cdot \rho^{rice} \\
CS &= \left[e \cdot \left(\sum_{i=1}^I \sum_{j=1}^J A_{ij} \cdot Y_{ij} \cdot \beta \cdot \sigma \right) + g \right] + \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} / \eta) \cdot A_{ij} \cdot v
\end{aligned} \tag{3}$$

The carbon footprint represents the total greenhouse gases (GHGs) released into the atmosphere as a result of human activities. Among these, nitrous oxide (N₂O) and methane (CH₄) are significant contributors to GHGs emissions. Research indicates that N₂O and CH₄ emissions can indirectly contribute to carbon emissions, making their inclusion essential in carbon footprint calculations [57]. To account for their impact, conversion factors of 298 for N₂O and 25 for CH₄ are applied, meaning that 1 kg of N₂O and 1 kg of CH₄ are equivalent to 298 kg and 25 kg of CO₂, respectively [58].

(1) Optimization goals of lower level decision makers

In the lower-level decision-making process, local management authorities, such as lower-level managers, need to coordinate and cooperate with farmers, who are decision-makers at a more microeconomic-level. At this level, decision-making reflects the priorities of regional authorities who represent the farmers as the key local stakeholders. These stakeholders are typically more concerned with direct economic returns and equitable outcomes (fairness), which directly influence production decisions and behavior. Therefore, when planning agricultural production, lower-level managers have to fully consider farmers' interests, balance the relationships between different stakeholders, optimize resource allocation, and improve agricultural efficiency.

Objective 1: Minimizing the Gini coefficient

At the subregional level, lower-level followers are concerned by their income level and income equity, which directly affects their production intentions and behavior [58]. Therefore, the objectives to focus on when planning agricultural production include income equity and total crop production [53]. The Gini coefficient, a statistical indicator reflecting the extent of the income distribution gap [61] can be used to assess the

fairness of the income distribution and guide farmers to develop more equitable agricultural production strategies [62]. In addition, lower-level followers have access to more granular data on farmers' individual income distributions and cropping cost structures, which makes the use of the Gini coefficient for within-group equity optimization practical [63]. While higher-level policymakers have access to aggregate resources across regions, they lack the support of microdata at the farm household level, and thus direct optimization of the Gini coefficient would not incorporate this information asymmetry. The objective function is expressed as follows:

$$\text{Min } F_{21} = \frac{1}{2I \sum_{i=1}^I \frac{TNB_i}{POP_i}} \sum_{a=1}^I \sum_{b=1}^I \left| \frac{TNB_a}{POP_a} - \frac{TNB_b}{POP_b} \right| \quad (4)$$

Objective 2: Maximizing agricultural food production

Before agricultural products enter the market, farmers often find it challenging to accurately estimate the market prices of these products because agricultural product prices are influenced by various factors, such as supply and demand, climatic conditions, and market environment. Therefore, farmers typically attempt to maximize total crop production to indirectly increase potential income. This objective can be expressed using the following mathematical formula:

$$\begin{aligned} \text{Max } F_{22} &= \sum_{i=1}^I FP_i \\ FP_i &= \sum_{j=1}^J A_{ij} Y_{ij} \end{aligned} \quad (5)$$

3.2.3 Constraints

(1) Minimum demand for water in each region.

Water allocation for agriculture is the process of optimally allocating and utilizing water resources based on the demand for agricultural production and the status of water supply in different sub-regions of the basin, which should be allocated to meet the minimum demand of different sub-regions of the basin, which is further bounded by the uncertain availability of water. This constraint is expressed as:

$$\sum_{j=1}^J A_{ij} \times \varepsilon_{ij} \times \chi_{ij} \leq DA_i \leq E^{Me}[Q(t)] \quad \forall i \quad (7)$$

(2) Food minimum demand constraint:

To ensure regional food security, the amount of agricultural production of food

should meet local minimum demand. This constraint is expressed as:

$$\sum_i^I FD_i \times POP_i \leq \sum_i^I \sum_j^J A_{ij} \times Y_{ij} \quad (8)$$

(3) *Land resource constraint:*

Generally, for the same crop in different regions and for different types of crops in the same region, the limitations on the extent of irrigated areas should be considered. Therefore, the area under crops should be within a certain range and this constraint can be expressed as:

$$\begin{aligned} A_{ij}^{\min} &\leq A_{ij} \leq A_{ij}^{\max} \\ \sum_{j=1}^J A_{ij} &\leq TK_i \end{aligned} \quad (9)$$

(4) *Minimum Economic Benefit Constraint.*

In order to promote sustainable development of the regional agricultural economy and increase the income of farmers, the output value of food crops should be higher than its economic efficiency index, which attracts farmers to grow food stably in order to ensure the development of the regional agricultural economy. The constraint can be expressed as follows:

$$\sum_{j=1}^J A_{ij} Y_{ij} MP_j \geq EB_i \quad \forall i \quad (10)$$

(5) *Nonpoint source pollution emission constraint:*

To protect the environmental quality of water bodies and reduce pollution from agricultural activities, it is important to consider and control the amount of nonpoint source pollutants. Impacts on the environmental level need to be considered and the amount of nonpoint source pollutants should be less than the maximum allowable total nitrogen loss. The constraint can be expressed as:

$$\sum_{i=1}^I \sum_{j=1}^J (NM \times q_{ij}^{fer-f} + q_{ij}^{fer-n} - NR_{ij}) \times A_{ij} \leq TN \times \sum_{i=1}^I TK_i \quad (11)$$

(6) *Non-negativity constraints*

The amounts of resources should not be negative and all decision variables should be non-negative, the constraint can be expressed as:

$$\begin{aligned} DA_i &\geq 0 \\ A_{ij} &\geq 0 \end{aligned} \quad (12)$$

3.2.4 Global Model

We define the global model in Equation (13). The global model presents the comprehensive integration of all previously described components—upper-level basin management decisions, lower-level regional government responses, fuzzy membership functions, and strategic interaction mechanisms—within a unified optimization framework.

The global model framework: 1) establishes the hierarchy between basin authorities and regional governments through a Stackelberg structure; 2) balances multiple objectives including economic benefits, resource efficiency, environmental impact, and social equity across decision levels; and 3) enforces all system constraints. This formulation enables systematic exploration of trade-offs while respecting cross-level strategic interactions, yielding balanced solutions for the MRB.

$$\begin{aligned}
\text{Max } F_{11} &= \sum_{i=1}^I TNB_i + ESV_i \\
\text{Min } F_{12} &= (ECO_2 + 298 \times EN_2O + 25 \times ECH_4 - CS) / \sum_{i=1}^I \sum_{j=1}^J A_{ij} Y_{ij} \\
\text{Max } F_{13} &= \sum_{i=1}^I \sum_{j=1}^J FP_{ij} / \sum_{i=1}^I (1 - \xi_i^{\text{loss}}) DA_i \\
&\left\{ \begin{aligned}
ESV_i &= f(\sum_{j=1}^J A_{ij}) \\
FP_{ij} &= A_{ij} \times Y_{ij} \\
TNB_i &= TB_i - TC_i \\
TB_i &= \sum_{j=1}^J A_{ij} Y_{ij} MP_j \\
\text{Min } F_{21} &= \frac{1}{2I \sum_{i=1}^I \frac{TNB_i}{POP_i}} \sum_{a=1}^I \sum_{b=1}^I \left| \frac{TNB_a}{POP_a} - \frac{TNB_b}{POP_b} \right| \\
\text{Max } F_{22} &= \sum_{i=1}^I FP_i \\
&\left\{ \begin{aligned}
TC_i &= \sum_{j=1}^J A_{ij} (Cn \times q^{\text{fer}-n} + Cf \times q^{\text{fer}-f} + Co \times q^{\text{fer}-o} + Cm \times q^{\text{fer}-m} \\
&\quad + Cp_{\text{pes}} \times q^{\text{pes}} + C_{\text{film}} \times q^{\text{film}} + C_{\text{des}} \times q^{\text{des}} + C_{\text{elec}} \times q^{\text{elec}}) \\
ECO_2 &= \sum_{i=1}^I \sum_{j=1}^J \left(A_{ij} \cdot \mu^{\text{fer}-n} \cdot q^{\text{fer}-n} + A_{ij} \cdot \mu^{\text{fer}-f} \cdot q^{\text{fer}-f} + A_{ij} \cdot \mu^{\text{fer}-o} \cdot q^{\text{fer}-o} + A_{ij} \cdot \mu^{\text{fer}-m} \cdot q^{\text{fer}-m} \right. \\
&\quad \left. + A_{ij} \cdot \mu^{\text{pes}} \cdot q^{\text{pes}} + A_{ij} \cdot \mu^{\text{film}} \cdot q^{\text{film}} + A_{ij} \cdot \mu^{\text{des}} \cdot q^{\text{des}} + A_{ij} \cdot \mu^{\text{elec}} \cdot q^{\text{elec}} \right) \quad (13) \\
EN_2O &= \sum_{i=1}^I \sum_{j=1}^J A_{ij} \cdot (ET_j + q^{\text{fer}-n} \cdot EN_j + q^{\text{fer}-f} \cdot EF_j) \\
ECH_4 &= \sum_{i=1}^I \sum_{j=1}^J A_{ij} \cdot \rho^{\text{rice}} \\
CS &= \left[e \cdot \left(\sum_{i=1}^I \sum_{j=1}^J A_{ij} \cdot Y_{ij} \cdot \beta \cdot \sigma \right) + g \right] + \sum_{i=1}^I \sum_{j=1}^J (Y_{ij} / \eta) \cdot A_{ij} \cdot v \\
FP_i &= \sum_{j=1}^J A_{ij} Y_{ij} \\
s.t. &\left\{ \begin{aligned}
\sum_{j=1}^J A_{ij} \times \varepsilon_{ij} \times \chi_{ij} &\leq DA_i \leq E^{\text{Me}}[Q(t)] \quad \forall i \\
\sum_i FD_i \times POP_i &\leq \sum_i \sum_j A_{ij} \times Y_{ij} \\
A_{ij}^{\min} &\leq A_{ij} \leq A_{ij}^{\max} \\
\sum_{j=1}^J A_{ij} &\leq TK_i \\
\sum_{j=1}^J A_{ij} Y_{ij} MP_j &\geq EB_i \quad \forall i \\
\sum_{i=1}^I \sum_{j=1}^J (NM \times q_{ij}^{\text{fer}-f} + q_{ij}^{\text{fer}-n} - NR_{ij}) \times A_{ij} &\leq TN \times \sum_{i=1}^I TK_i \\
A_{ij}, DA_i &\geq 0
\end{aligned} \right.
\end{aligned}
\right.
\end{aligned}$$

3.3 Model solution

3.3.1 Objective conversion

To effectively handle the uncertainty of regional water availability, while optimizing multiple competing objectives at different decision levels, our global model employs a sophisticated methodological approach combining distance functions and fuzzy set theory [41,67]. Building on the literature, we focus on uncertainty of water

availability, since water resources are the dominant driver of crop-yield fluctuations and therefore represents the most critical source of uncertainty in the WEFC nexus [73, 74, 75, 76], largely due to climate change. Indeed, climate change has induced a series of adverse impacts, including river droughts, precipitation variability, and enhanced rainfall evaporation, which result in uncertainties regarding available water resources and other water sources within watersheds. Furthermore, uncertainties in external water sources also contribute to the complexity and uncertainty of decision-making processes by watershed management authorities [77,78]. Therefore, we characterize water availability, the primary source of uncertainty, as a discrete fuzzy random variable. Based on historical hydrological data, we define several water availability scenarios over which there is a probability distribution. These fuzzy variables are then converted into a single, deterministic expected value of available water for use in the model's constraints. This is achieved through a fuzzy expected value calculation [81, 85], which provides a computationally tractable means of incorporating uncertainty into the optimization. The detailed procedure is provided in Appendix B of the supplementary materials.

Using the concept of a distance function to convert multiple conflicting objectives in a bi-level optimization problem into two objectives is a common approach to solving such models as previously outlined [16,41,42]. This allows the model to be reduced to a single-objective optimization problem, from which we can obtain the Pareto-optimal frontier solution using fuzzy set theory [43]. The fuzzy approach enables systematic quantification of satisfaction degrees for each objective under varying uncertain conditions, transforming deterministic constraints into flexible fuzzy constraints that accommodate parameter variability.

To obtain more detailed and reliable modeling results, this study continues to use the concept of distance function to transform the bi-level multi-objective planning problem into a single-objective solution [45,46]. This transformation process enables balanced consideration of competing priorities across decision levels while systematically accounting for system uncertainties. Specifically, the proposed computational steps are shown below:

(1) Modeling bi-level multi-objective nonlinear planning for water-energy-food-carbon nexus in agricultural systems;

(2) The maximum and minimum values of the five objective functions in the upper and lower levels are denoted as: $F_{11}^{\max}$, $F_{12}^{\max}$, $F_{13}^{\max}$, $F_{21}^{\max}$, $F_{22}^{\max}$, $F_{11}^{\min}$, $F_{12}^{\min}$, $F_{13}^{\min}$, $F_{21}^{\min}$, $F_{22}^{\min}$;

(3) Convert multiple objective functions in each layer into a single objective function;

for upper level:

$$F_{upper} = \frac{F_{11}^{\max} - F_{11}}{F_{11}^{\max} - F_{11}^{\min}} + \frac{F_{12} - F_{12}^{\min}}{F_{12}^{\max} - F_{12}^{\min}} + \frac{F_{13} - F_{13}^{\min}}{F_{13}^{\max} - F_{13}^{\min}} \quad (13)$$

for lower level:

$$F_{lower} = \frac{F_{21}^{\max} - F_{11}}{F_{21}^{\max} - F_{21}^{\min}} + \frac{F_{22} - F_{22}^{\min}}{F_{22}^{\max} - F_{22}^{\min}} \quad (14)$$

(4) Solve for the decision variables and objective function values F'_{ufu} using the upper level objective F_{upper} . Bring the decision variables into the lower level and solve for the lower level objective function values F'_{lfu} . Solve for the decision variables and objective function values F'_{lfl} using the lower level objective F_{lower} . Bring the decision variables into the upper level and solve for the upper level objective function values F'_{ufl} [64,65].

(5) Construct membership functions for upper and lower level

for upper level:

$$G_1(F_{upper}) = \begin{cases} 1 & ,if F_{upper} < F'_{ufu} \\ \frac{F_{upper} - F'_{ufu}}{F'_{ufl} - F'_{ufu}} & ,if F'_{ufu} < F_{upper} < F'_{ufl} \\ 0 & ,if F_{upper} > F'_{ufl} \end{cases} \quad (15)$$

for lower level:

$$G_2(F_{lower}) = \begin{cases} 1 & ,if F_{lower} < F'_{lfl} \\ \frac{F_{lower} - F'_{lfl}}{F'_{lfu} - F'_{lfl}} & ,if F'_{lfl} < F_{lower} < F'_{lfu} \\ 0 & ,if F_{lower} > F'_{lfu} \end{cases} \quad (16)$$

Transforming a bi-level optimization model into a new objective optimization model:

$$\max \alpha \cdot G_1(F_{upper}) + (1 - \alpha) \cdot G_2(F_{lower}), \quad \alpha \in [0,1] \quad (17)$$

3.3.2 Interactive Genetic Algorithms for Elite Strategies for Undominated Sorting

According to the above model transformation, this paper proposes an interactive non-dominated sorted elite genetic algorithm based on interaction to solve the model, the specific pseudo-code is shown in the supplementary material (Appendix A), the flow chart is shown in Fig. 3, and the relevant computational steps are as follows:

Step 1: Considering the global resource allocation context, each transformed objective function is assigned a corresponding weight coefficient. These weights reflect the relative importance of each objective in the overall optimization problem.

Step 2: Initial Population Generation. Within the feasible solution space defined by the problem constraints, an initial population of candidate solutions is generated randomly. The following parameters govern this process:

- N : Population size.
- G_{max} : Maximum number of generations (iterations).
- P_c : Crossover probability.
- P_m : Mutation probability.

Resource constraints are defined as $g_k(x) \leq 0, k = 1, 2, \dots, K$. The initial population P_0 consists of N randomly generated individuals that satisfy these constraints.

Step 3: Iterative Optimization ($t = 0, 1, \dots, G_{max} - 1$)

Each generation of the algorithm comprises the following steps:

Non-dominated Sorting: The population P_t is subjected to non-dominated sorting, resulting in a set of non-dominated fronts $F_{11}, F_{12}, \dots$

Crowding Distance Calculation: The crowding distance, a measure of solution density, is computed for each individual within each non-dominated front.

Interactive Weight Elicitation and Candidate Set Construction:

- **Pareto Front Approximation Visualization:** The first non-dominated front (F_{11}), representing an approximation of the Pareto front, is visualized and presented to the user.
- **Weight Input:** The user is prompted to provide weights (w_m) for each objective function ($m = 1, 2, \dots, M$), subject to the constraints $w_m \geq 0$ and $\sum_{m=1}^M w_m = 1$.
- **Candidate Set Formation:** Solutions from F_{11} are copied to a candidate set C . If the cardinality of $|C|$ is less than a predefined threshold T , solutions from subsequent fronts ($F_{12}, F_{13}, \dots$) are sequentially added to C until $|C| = T$ or all fronts have been considered.

Fitness Evaluation (for Candidate Set C): For each individual x_i in C , a fitness value $fitness(x_i)$ is calculated. A weighted sum approach, incorporating normalized objective function values, is employed: $fitness(x_i) = \sum_{m=1}^M w_m \cdot \frac{f_m^{max} - f_m(x_i)}{f_m^{max} - f_m^{min}}$, where f_m^{max} and f_m^{min} represent the maximum and minimum values, respectively, of the m -th objective function within C .

Genetic Operators:

- **Selection:** Parent individuals are selected based on their $fitness(x_i)$.
- **Crossover:** Crossover is applied to selected parents with probability P_c to generate offspring.
- **Mutation:** Mutation is applied to offspring with probability P_m .

Environmental Selection and Elitism: The parent population P_t and the offspring population are combined. N individuals are selected from this combined population to form the next generation P_{t+1} , utilizing non-dominated sorting and crowding distance comparison to maintain elitism and diversity.

Step 4. Termination. After G_{max} iterations, the algorithm terminates. The final candidate set C is returned, and the user is prompted to provide final weights. The solution from C with the highest fitness value, based on these final weights, is selected as the recommended solution.

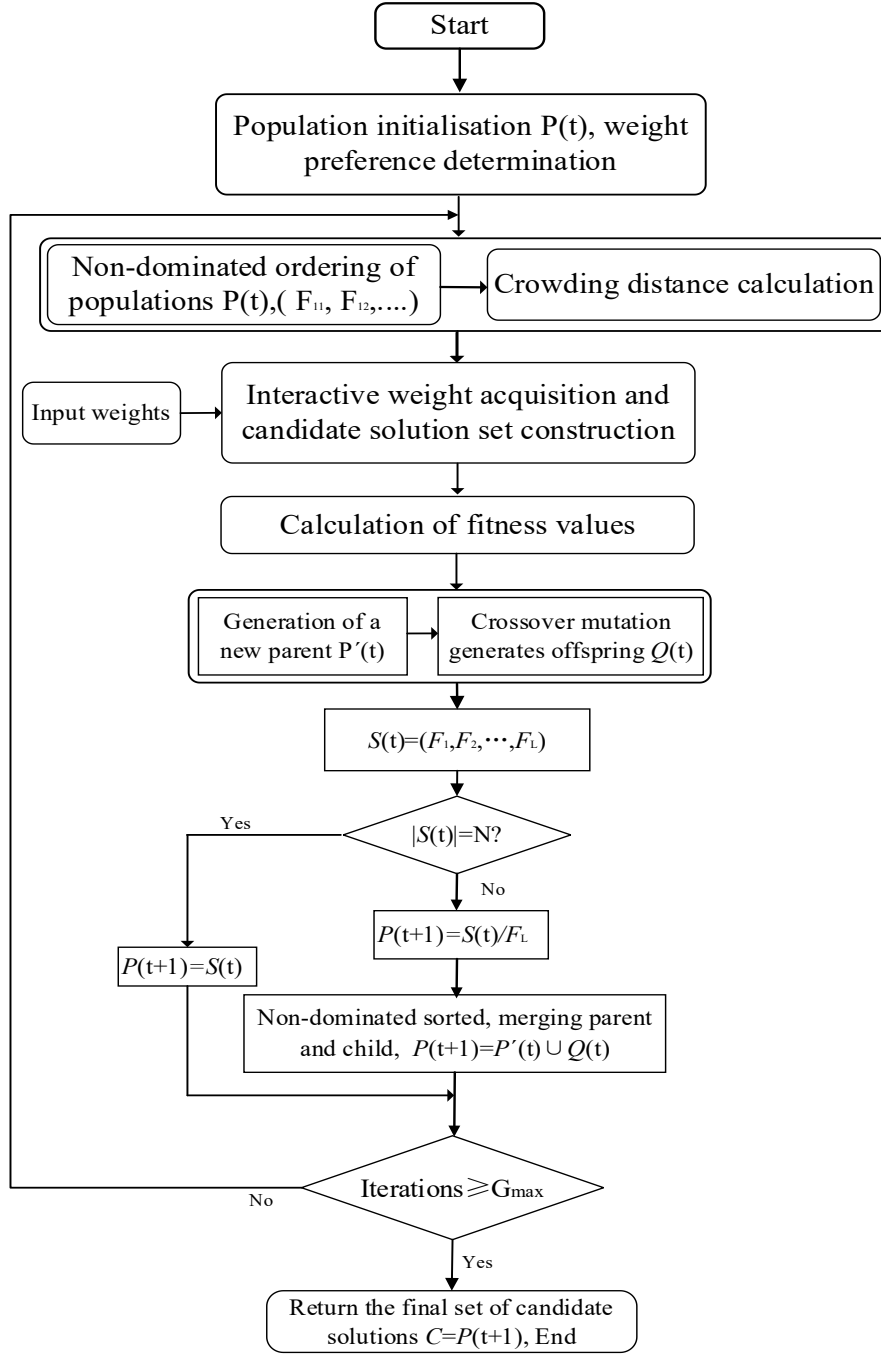


Fig. 3. Flowchart of genetic algorithm for non-dominated sorting elite strategy

4. Results & discussion

This study establishes a bi-level decision-making framework for water and land resource optimization allocation frameworks (Fig.2), aiming to achieve joint optimization of objectives at both upper and lower levels. The upper-level decision-makers (basin management authorities) primarily focus on maximizing economic benefits (F_{11}), maximizing irrigation water productivity (F_{12}), and minimizing carbon emissions per unit of food production (F_{13}), while lower-level decision-makers

(regional government departments) emphasize minimizing the Gini coefficient (F_{21}) and maximizing agricultural food production (F_{22}). Based on the fuzzy membership method, this study transforms the bi-level objective functions into a single objective function, introducing weight coefficients [0.1, 0.2, ..., 0.9] to characterize the relative importance between upper and lower decision levels. With nine weight combinations, ranging from (0.1,0.9), (0.2,0.8), ..., to (0.9,0.1), we systematically analyze resource allocation effects under different decision preferences. An increase in weights indicates enhanced emphasis on the corresponding decision level's objectives, with higher upper-level weights favoring economic and environmental objectives, while higher lower-level weights emphasize equity and food security.

4.1 Analysis of the results of the objective function

The data results in Table 7 indicate differentiated response characteristics for each objective function under varying weights, revealing complex interactions between decision levels and objectives.

For upper-level decision objectives, the economic benefit (F_{11}) exhibited fluctuations as weights changed from (0.1,0.9) to (0.9,0.1), ranging from $[287.706 \times 10^8 \text{ RMB to } 289.650 \times 10^8 \text{ RMB}]$. Specifically, it reached $289.225 \times 10^8 \text{ RMB}$ at (0.1,0.9), decreased to minimum $287.706 \times 10^8 \text{ RMB}$ at (0.2,0.8), stabilized at $288.438 \times 10^8 \text{ RMB}$ when weights were (0.4,0.6) and (0.5,0.5), and reached a high of $289.250 \times 10^8 \text{ RMB}$ at (0.7,0.3). **This non-monotonic response highlights the model's sensitivity to decision-maker preferences and reveals the intricate, non-linear trade-offs between economic, environmental, and social objectives within the WEFC nexus.** Maximizing irrigation water productivity (F_{12}) ranged from $[1.655 \text{ kg/m}^3, 1.694 \text{ kg/m}^3]$, showing an overall declining trend from 1.694 kg/m^3 at (0.1,0.9) to 1.655 kg/m^3 at (0.8,0.2) and (0.9,0.1), with intermediate values of 1.657 kg/m^3 and 1.656 kg/m^3 at (0.4,0.6) and (0.5,0.5) respectively. Minimizing carbon emissions per unit of food production (F_{13}) fluctuated within $[5.649 \text{ kg CO}_2\text{e}, 5.758 \text{ kg CO}_2\text{e}]$ as the weights changed. Initially, emissions decreased from $5.661 \text{ kg CO}_2\text{e}$ to a minimum of $5.649 \text{ kg CO}_2\text{e}$ as the weights shifted from (0.1, 0.9) to (0.2, 0.8). Subsequently, they increased gradually to $5.730 \text{ kg CO}_2\text{e}$ when weights moved from (0.3, 0.7) to (0.5, 0.5). Emissions then continued to rise, reaching a maximum of $5.758 \text{ kg CO}_2\text{e}$ at (0.8, 0.2), before finally decreasing to $5.714 \text{ kg CO}_2\text{e}$ at (0.9, 0.1). Again, from the nonlinear response, this demonstrates the complexity of modelling a highly complex environment.

Regarding lower-level decision objectives, resource allocation equity, measured

by the Gini coefficient (F_{21}), displayed significant fluctuations within [0.193, 0.213]. The coefficient started at 0.213 (0.1,0.9), decreased to 0.201 at (0.4,0.6), reached 0.200 at (0.5,0.5), dropped to 0.196 at (0.6,0.4), increased to 0.207 at (0.7,0.3), and finally decreased to 0.193 and 0.198 at (0.8,0.2) and (0.9,0.1) respectively. In contrast, food production (F_{22}) demonstrated remarkable stability, fluctuating minimally within $[989.820 \times 10^4 \text{kg}, 989.836 \times 10^4 \text{kg}]$. Production levels remained nearly constant across different weight combinations, starting at $989.821 \times 10^4 \text{kg}$ at (0.1,0.9), reaching a peak of $989.836 \times 10^4 \text{kg}$ at (0.2,0.8), and maintaining $989.821 \times 10^4 \text{kg}$ at both (0.5,0.5) and (0.9,0.1), indicating strong model robustness in terms of food production stability. We also observe clearly non-monotonic fluctuations in the lower-level decision objectives as weights change, highlighting the inherent complexity in decision-making.

Our analysis identifies the (0.5,0.5) weight combination, representing equal preference for upper- and lower- level objectives, as a point of optimal system balance. Our estimates show that economic benefits maintain a relatively high level of 288.438×10^8 RMB, with grain production efficiency at 1.656kg/m^3 , carbon emissions controlled at a moderate level of $5.730 \text{ kg CO}_2\text{e}$, resource distribution Gini coefficient at 0.200, and stable food production at $989.821 \times 10^4 \text{kg}$. Under this balanced weighting, the model yields an allocation scheme that achieves a balance among economic benefits, resource distribution equity, and environmental protection, providing scientific support for agricultural water and land resource optimization in the MRB. The results demonstrate that balanced consideration of both upper and lower-level objectives can lead to sustainable and equitable resource allocation outcomes.

Tab 7 Objective function values under different weighting scenarios

Objectives	Weight (Upper, Lower)	(0.1,0.9)	(0.2,0.8)	(0.3,0.7)	(0.4,0.6)	(0.5, 0.5)
Upper level objective	F_{11} (RMB)	289.225×10^8	287.706×10^8	287.949×10^8	288.438×10^8	288.438×10^8
	F_{12} (kg/m^3)	1.694	1.663	1.659	1.657	1.656
	F_{13} ($\text{kg CO}_2\text{e}$)	5.661	5.649	5.689	5.712	5.730
Lower level objective	F_{21} (Gini)	0.213	0.205	0.202	0.201	0.200
	F_{22} (kg)	989.821×10^4	989.836×10^4	989.824×10^4	989.828×10^4	989.821×10^4
	Weight	(0.6,0.4)	(0.7,0.3)	(0.8,0.2)	(0.9,0.1)	
Upper level objective	F_{11} (RMB)	288.052×10^8	289.250×10^8	288.346×10^8	288.074×10^8	
	F_{12} (kg/m^3)	1.657	1.656	1.655	1.655	
	F_{13} ($\text{kg CO}_2\text{e}$)	5.711	5.730	5.758	5.714	
Lower level objective	F_{21} (Gini)	0.196	0.207	0.193	0.198	
	F_{22} (kg)	989.820×10^4	989.821×10^4	989.824×10^4	989.821×10^4	

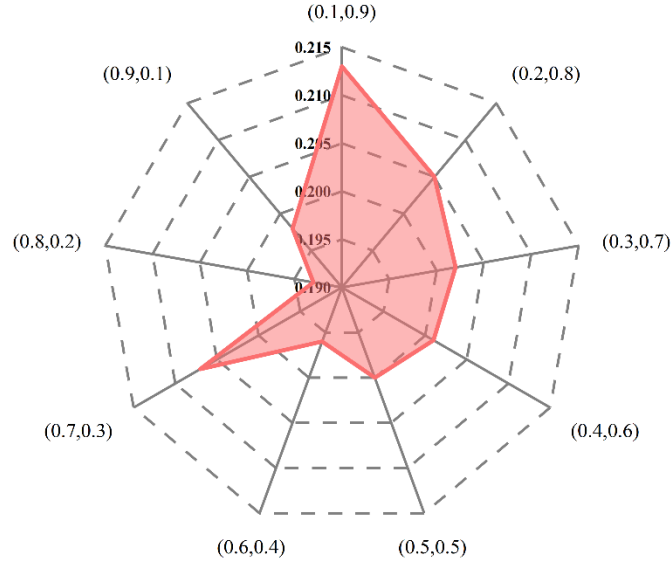


Fig.4 Gini coefficient values under different weights

4.2 Agricultural water resource allocation results

Table 8 and Fig.5 presents the agricultural water resource allocation results across different regions of the MRB under varying weights, revealing significant spatial heterogeneity. The water resource allocation forms a distinct "three-tier gradient" pattern across regions: The first tier is represented by Chengdu, with allocations fluctuating between $[32.9071 \times 10^8 \text{m}^3, 32.9123 \times 10^8 \text{m}^3]$, accounting for 41.8% of the total allocation. The second tier comprises Leshan $[10.1376 \times 10^8 \text{m}^3, 10.1389 \times 10^8 \text{m}^3]$ (12.9%) and Yibin $[8.5990 \times 10^8 \text{m}^3, 8.6054 \times 10^8 \text{m}^3]$ (10.9%). The third tier consists of Meishan $[7.5943 \times 10^8 \text{m}^3, 7.6020 \times 10^8 \text{m}^3]$, Neijiang $[6.2409 \times 10^8 \text{m}^3, 6.2441 \times 10^8 \text{m}^3]$, Ya'an $[4.9798 \times 10^8 \text{m}^3, 4.9849 \times 10^8 \text{m}^3]$, and Zigong $[4.4193 \times 10^8 \text{m}^3, 4.4226 \times 10^8 \text{m}^3]$, accounting for 9.7%, 7.9%, 6.3%, and 5.6% of the total allocation respectively. Aba Prefecture receives the smallest allocation $[1.5288 \times 10^8 \text{m}^3, 1.5378 \times 10^8 \text{m}^3]$, representing only 2.0% of the total.

From a spatiotemporal perspective, as the upper-level decision weight varies from 0.1 to 0.9, different regions exhibit varying stability characteristics. Chengdu demonstrates the highest stability with a relative fluctuation rate of only 0.016% (maximum difference of $0.0052 \times 10^8 \text{m}^3$), followed by Leshan and Yibin with fluctuation rates of 0.013% and 0.074% respectively. In contrast, Aba Prefecture shows the highest sensitivity to weight changes with a relative fluctuation rate of 0.586%, reflecting the adaptive characteristics of water resource allocation across regions.

These spatial differences can be attributed to significant variations in regional

resource endowments and agricultural development levels. Chengdu possesses 33.2×10^4 hectares of arable land (13.01% of the province's cultivable land) with favorable climatic conditions, resulting in high agricultural water demand. Conversely, Aba Prefecture, constrained by high altitude and vertical climate zones, has only 6.58×10^4 hectares of arable land, predominantly practicing dry farming with relatively lower water resource demands. This efficiency-based allocation strategy effectively matches water resource utilization with regional agricultural development needs.

In this study, the weight combination (0.5, 0.5) was selected as the resource allocation plan, which consider the strategic interaction between decision-makers who have different preferences, while avoiding overly conservative outcomes. Under this strategy, agricultural water resources are further optimized: Chengdu maintains a high level at $32.9123 \times 10^8 \text{m}^3$, while Leshan ($10.1388 \times 10^8 \text{m}^3$) and Yibin ($8.5990 \times 10^8 \text{m}^3$) form the core area of water resource utilization, collectively accounting for 65.6% of total allocation. Meishan ($7.5963 \times 10^8 \text{m}^3$), Neijiang ($6.2418 \times 10^8 \text{m}^3$), Ya'an ($4.9848 \times 10^8 \text{m}^3$), and Zigong ($4.4218 \times 10^8 \text{m}^3$) constitute the secondary utilization zone, representing 29.5% of the total allocation. Aba ($1.5378 \times 10^8 \text{m}^3$) exhibits characteristic low-intensity utilization patterns. This differentiated allocation pattern reflects both the actual demands of regional agricultural development and the spatial optimization of water resource utilization efficiency.

Tab. 8. Results of water allocation in MRB under different weighting preferences ($10 \times 10^8 \text{m}^3$)

Subarea	(0.1, 0.9)	(0.2, 0.8)	(0.3, 0.7)	(0.4, 0.6)	(0.5, 0.5)	(0.6, 0.4)	(0.7, 0.3)	(0.8, 0.2)	(0.9, 0.1)
Chengdu	32.9115	32.9107	32.9071	32.9089	32.9123	32.9122	32.9121	32.9123	32.9123
Zigong	4.4222	4.4193	4.4199	4.4220	4.4218	4.4226	4.4226	4.4226	4.4226
Neijiang	6.2439	6.2435	6.2438	6.2409	6.2418	6.2441	6.2440	6.2440	6.2441
Leshan	10.1383	10.1389	10.1376	10.1388	10.1388	10.1388	10.1389	10.1387	10.1389
Meishan	7.6020	7.6015	7.5943	7.6008	7.5963	7.6019	7.6020	7.6019	7.6020
Yibin	8.6053	8.6052	8.6046	8.6054	8.5990	8.6054	8.6053	8.6045	8.6054
Ya'an	4.9849	4.9841	4.9845	4.9798	4.9848	4.9847	4.9842	4.9848	4.9849
Aba	1.5377	1.5310	1.5378	1.5288	1.5378	1.5375	1.5378	1.5375	1.5378

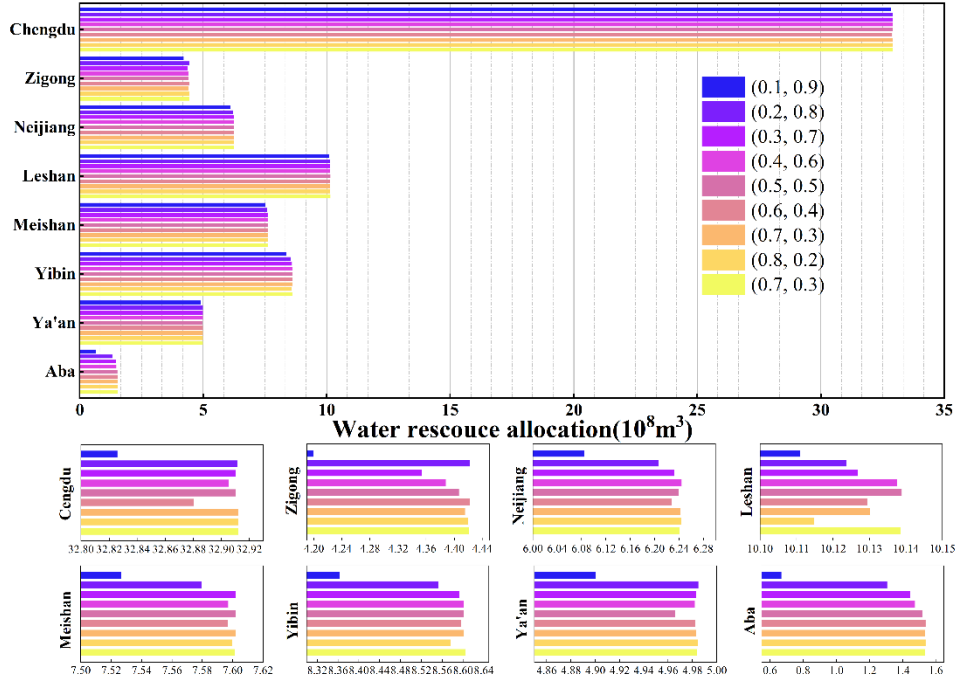


Fig.5 Water resource allocation scheme uncertainty in subarea

4.3 Agricultural land resource allocation results

Based on the water and land resource endowment characteristics of the MRB, this study analyzes the spatial distribution patterns of crop planting structures under different weights. The results of spatial allocation of water and land resources under different weight changes are given in Fig. 6, which shows that decision-makers' preferences affect the pattern of water and land resources allocation in MRB. However, data in Table 9 reveals significant regional differences in planting structures, particularly reflected in rice cultivation areas substantially exceeding other crop areas. Additionally, as the weights change, water resource allocation shows distinct regional differences: The Chengdu Plain consistently maintains high water resource allocation quotas, aligning with its status as a regional agricultural core area; midstream regions (such as Meishan and Neijiang) show relatively stable water resource allocation characteristics; while upstream mountainous areas (such as Aba) receive less water allocation with minimal fluctuations. This water resource allocation pattern closely correlates with land resource utilization, subsequently influencing regional planting structure formation. These differences primarily stem from two aspects: first, the unique subtropical climate characteristics of the MRB, where warm and humid climate conditions, abundant precipitation, and fertile land provide superior natural foundations for rice cultivation [22,51]; second, the heterogeneity in agricultural development levels

and resource endowments across regions leads to spatial heterogeneity in planting structures.

Similarly, at the weight combination (0.5, 0.5), the system reaches a potentially optimal balance state, with upstream regions (Aba and Chengdu) showing distinct planting structure contrasts. The Chengdu Plain is dominated by rice cultivation ($233.103 \times 10^3 \text{ha}$), with wheat and corn areas of $102.773 \times 10^3 \text{ha}$ and $44.463 \times 10^3 \text{ha}$ respectively, forming a rice-led diversified grain crop planting pattern. In contrast, Aba Prefecture exhibits typical dry farming characteristics, primarily growing wheat ($14.960 \times 10^3 \text{ha}$) and corn ($15.794 \times 10^3 \text{ha}$), with minimal rice cultivation ($0.908 \times 10^3 \text{ha}$). This differentiated pattern primarily reflects the constraints of topographical conditions and climate characteristics.

The middle and downstream regions (Ya'an, Meishan, Neijiang, Leshan, Zigong, and Yibin) display more complex spatial differentiation characteristics. Yibin has developed a balanced grain crop development pattern dominated by rice ($150.663 \times 10^3 \text{ha}$), with wheat and corn areas reaching $78.660 \times 10^3 \text{ha}$ and $69.932 \times 10^3 \text{ha}$ respectively. Meishan, Neijiang, and Zigong exhibit similar planting structures, predominantly following a rice-wheat pattern, with Meishan's rice and wheat areas approximately $110.791 \times 10^3 \text{ha}$ and $76.098 \times 10^3 \text{ha}$; Neijiang approximately $83.949 \times 10^3 \text{ha}$ and $69.744 \times 10^3 \text{ha}$; and Zigong approximately $81.274 \times 10^3 \text{ha}$ and $47.969 \times 10^3 \text{ha}$. Notably, Leshan and Ya'an have formed unique planting patterns: Leshan primarily grows rice ($89.258 \times 10^3 \text{ha}$) and corn ($49.70 \times 10^3 \text{ha}$), while Ya'an shows balanced characteristics with comparable rice ($29.818 \times 10^3 \text{ha}$) and corn ($28.810 \times 10^3 \text{ha}$) cultivation areas.

Combining the spatial dynamic changes of water and land resource allocation in Fig. 6 with the land resource allocation scheme in Table 9, the MRB's agricultural resource allocation exhibits characteristics consistent with (i) determining land use by water availability and, (ii) adapting crops to land conditions. This allocation pattern considers both water resource constraints and rational regional agricultural development layout. The optimized planting structure fully considers regional differences in hydrothermal conditions and land resources, achieving efficient utilization of agricultural resources and providing scientific basis for regional agricultural sustainable development.

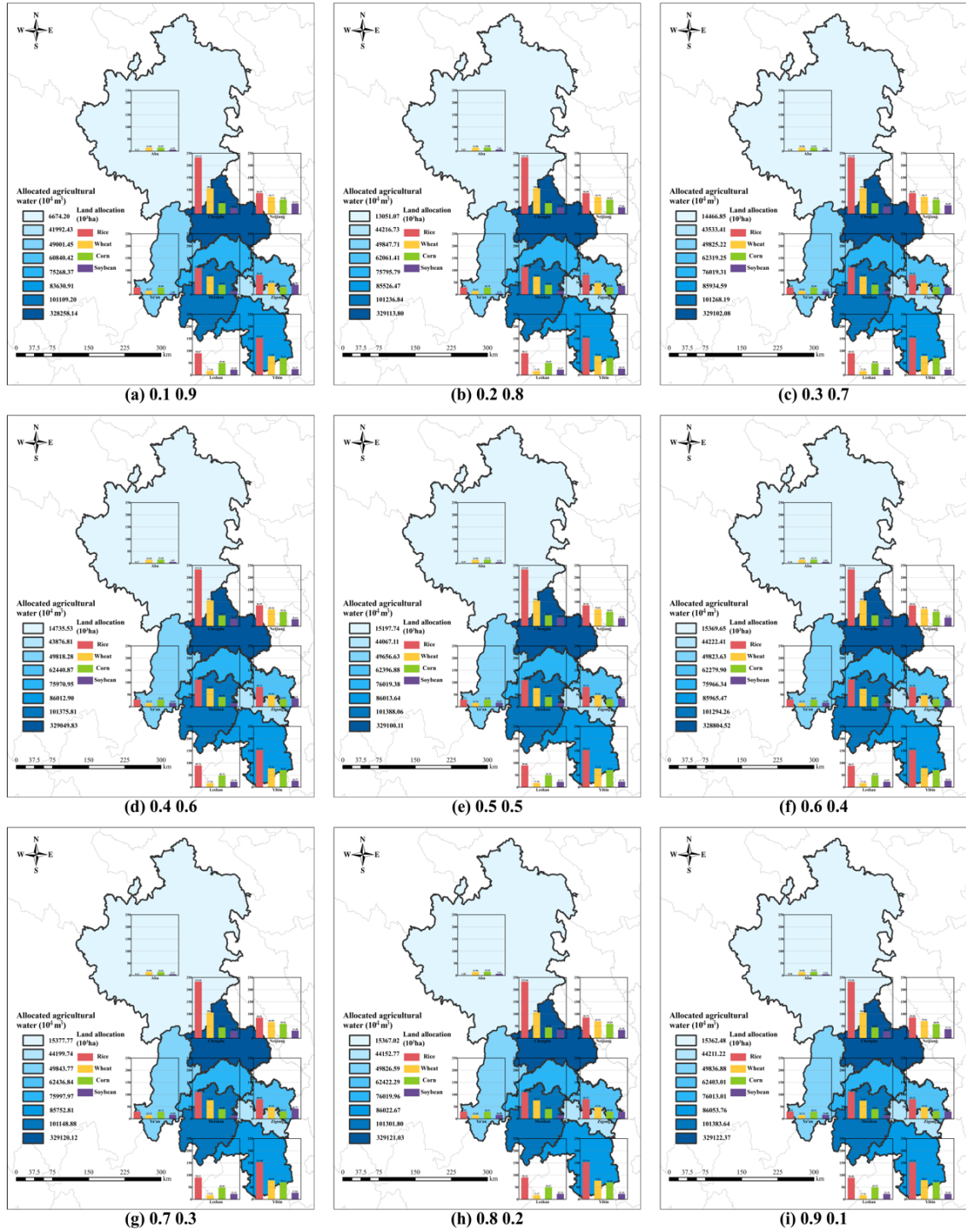


Fig.6 Water-land resource allocation under different weighting values in the MRB

Table 9 Land resource allocation scheme at (0.5,0.5) (10³ha)

Subarea	Rice	Wheat	Corn	Soybean
Chengdu	233.103	102.773	44.463	36.432
Zigong	81.274	47.696	30.923	35.042
Neijiang	83.949	69.744	58.480	43.731
Leshan	89.258	17.520	49.700	22.758
Meishan	110.791	76.098	39.741	14.354
Yibin	150.663	78.660	69.932	25.777
Yaan	29.818	15.538	28.810	15.748
Aba	0.908	14.960	15.794	5.671

4.4 Analysis of carbon emissions from agricultural cultivation

Based on agricultural carbon emission measurements across eight cities in the MRB, this study conducts quantitative analysis across dimensions including total emissions, compositional characteristics, and regional differences to assess the environmental impact of agricultural activities. Results in Table 10 reveal significant regional heterogeneity and source diversity in agricultural carbon emissions, characteristics linked to regional physiographic conditions, agricultural structure, and technological factors.

Based on the comprehensive analysis of net carbon emissions and their regional distribution characteristics, the study area can be categorized into three hierarchical gradients: high-emission zones (Chengdu City: 161.46×10^8 kg; Meishan City: 149.62×10^8 kg), medium-emission zones (Leshan City: 74.90×10^8 kg; Neijiang City: 69.72×10^8 kg; Yibin City: 78.62×10^8 kg), and low-emission zones (Ya'an City: 50.38×10^8 kg; Zigong City: 50.23×10^8 kg; Aba Prefecture: 6.53×10^8 kg). This distribution pattern is closely related to the regional agricultural development level, land use intensity, and topographical and geomorphological characteristics. In particular, the Chengdu Plain region, with its high degree of intensive agriculture, advanced agricultural modernization, and large cultivated land area, has significantly higher total carbon emissions compared to the mountainous and hilly regions. Notably, Chengdu City, as the regional agricultural development center, has a carbon emission of 161.46×10^8 kg, far exceeding the regional average, highlighting the close correlation between agricultural development and carbon emissions.

A multidimensional analysis of the sources of carbon emissions reveals that the composition of agricultural carbon emissions exhibits distinct hierarchical and regional characteristics, specifically: (i) Indirect carbon emissions from CO dominate in all cities,

generally accounting for more than 80%. Meishan City and Chengdu City have emission levels of 142.93×10^8 kg and 136.97×10^8 kg, respectively, reflecting the high intensity of fertilizer application. It is particularly noteworthy that in Ya'an City, this proportion reaches as high as 99.10%, indicating that local agricultural production is excessively dependent on fertilizer input, and nitrogen fertilizer management urgently needs optimization. This high proportion of indirect carbon emissions from CO not only affects environmental quality but also reflects the unsustainability of agricultural production methods. (ii) Indirect carbon emissions from CH₄ exhibit significant regional differences, with proportions ranging from 1.53% to 20.52%. Rice-growing areas (such as Yibin City) have a higher proportion of CH₄ emissions, reaching 20.52%, which is related to the large rice cultivation area and long history of rice planting in the region. In contrast, CH₄ emissions in alpine regions (such as Aba Prefecture) account for only 1.53%, reflecting the regional characteristics of agricultural structure and cultivation systems. (iii) Although direct carbon emissions account for a relatively small proportion (1.46%-5.21%), there are obvious differences between regions. Chengdu City, with its higher level of agricultural mechanization and frequent use of farmland irrigation and agricultural machinery, has direct carbon emissions of 12.99×10^8 kg, significantly higher than other regions. This phenomenon indicates that the process of agricultural modernization also entails an environmental cost of increased carbon emissions.

Analyzing the regional differences in carbon sequestration effects reveals that the carbon sequestration capacity of the MRG has distinct geographical gradient characteristics. Aba Prefecture, benefiting from its unique alpine ecosystem characteristics and lower agricultural development intensity, has a carbon sequestration rate as high as 27.72%, significantly higher than the plain agricultural areas. This high carbon sequestration rate is mainly derived from the region's abundant grassland resources and good vegetation cover conditions. In contrast, Leshan City has a carbon sequestration rate of only 3.81%, reflecting the relatively weak carbon sink function of the ecosystem in intensive agricultural regions. These regional differences in carbon sequestration effects provide an important basis for formulating differentiated carbon emission reduction policies.

Table 10 Characteristics of Agricultural Carbon Emissions in MRB (10^8kg)

Subarea	Direct carbon emissions	Indirect carbon emissions from CO	Indirect carbon emissions from CH ₄	Carbon sequestration	Net carbon emissions
Chengdu	12.99	136.97	24.53	13.03	161.46
Zigong	2.10	48.33	8.57	8.77	50.23
Neijiang	2.54	62.22	9.04	4.08	69.72
Leshan	2.03	66.36	9.37	2.85	74.90
Meishan	2.19	142.93	11.78	7.28	149.62
Yibin	3.11	66.54	16.14	7.18	78.62
Ya'an	1.30	49.93	3.14	4.00	50.38
Aba	0.34	7.90	0.10	1.81	6.53

4.5 Discussion

This study explored agricultural water and land resource optimization allocation under the WEFC nexus framework through a bi-level multi-objective programming model. The results not only validate the model's effectiveness but also reveal key scientific issues regarding resource allocation efficiency, spatial heterogeneity, and carbon emission control. We discuss these findings across three dimensions.

First, the findings highlight the characteristics of dynamic equilibrium in resource allocation efficiency. The minimal fluctuations in economic benefits (0.54%) and food productivity (2.36%) despite significant shifts in decision-maker weights suggest the system can achieve Pareto-like improvements, where reallocating resources to meet one objective need not compromise others. This finding highlights the resilience achievable through multi-objective optimization. Additionally, when there is an equal weighting between upper- and lower-level objectives, the system achieves an optimal balance, ensuring maximum water resource allocation for Chengdu and enabling rice cultivation to fully capitalize on economies of scale. This underscores the limitations of traditional, spatially-uniform allocation policies. Our results demonstrate that an efficiency-driven allocation, which tailors resource distribution to regional capacity and demand, is most beneficial for the entire system. This provides robust, model-driven evidence for policymakers to use more tailored schemes and policies that account for both basin-level and regional-level preferences across different regions, consistent with Liu et al. [12].

Second, our results identify pronounced spatial heterogeneity in agricultural resource allocation, revealing a distinct “three-tier gradient” pattern across the Chengdu Plain, middle-stream regions (Leshan, Yibin), and upstream mountainous areas (Aba

Prefecture). For example, the model allocates water to the highly productive Chengdu Plain at a rate sufficient for rice cultivation that is over 250 times greater than that for the arid, mountainous Aba Prefecture. This underscores the limitations of traditional allocation approaches; we demonstrate that an efficiency-driven allocation which tailors resource distribution to regional capacity can better optimize overall system performance by aligning resource distribution with regional capacities and demands. These findings provide robust evidence for policymakers, suggesting that tailored allocation frameworks that consider both environmental and economic factors are required to achieve sustainable and equitable resource management. Compared to Chen et al. [43] and accompanying literature, our findings also suggest a further insight that spatial heterogeneity in resource allocation is influenced not only by natural conditions but also by socioeconomic development levels, for example, comparing a highly economically developed region like the Chengdu Plain with, a less developed area such as the Aba Prefecture, suggesting differences may go beyond environmental factors.

Third, our study provides valuable insights into the multi-source and complex nature of agricultural carbon emissions, revealing a clear “three-tier gradient” that mirrors regions' agricultural productivity and economic development, challenging oversimplified allocation policies that are not region-based, with high-emission areas (Chengdu and Meishan), medium-emission areas (Leshan, Neijiang, and Yibin), and low-emission areas (Ya'an, Zigong, and Aba Prefecture) with indirect CO being the primary source of emissions. There are also significant regional variations in CH₄ emissions—particularly higher in rice cultivation areas like Yibin (20.52%) and lower in high-altitude regions such as Aba (1.53%). This underscores the need for targeted mitigation strategies, highlighting the influence of agricultural practices in different locations. For example, this finding implies that basin-wide carbon reduction targets should be disaggregated into region-specific strategies that account for local cropping systems (e.g., CH₄ mitigation in rice-dominant areas vs. N₂O management in fertilizer-intensive zones). Furthermore, the marked differences in carbon sequestration potential, with Aba Prefecture achieving a sequestration rate of 27.72% compared to only 3.81% in Leshan, further emphasize the importance of region-specific approaches to emissions management. Compared to other studies like Feng et al. [22], these findings allow us to more precisely quantify regional differences in agricultural-related emissions within the Water-Energy-Food-Carbon (WEFC) nexus, emphasizing the need for tailored emission reduction strategies while balancing productivity and environmental sustainability [66].

5. Conclusions and policy recommendation

With intensifying global climate change and continuous population growth, agricultural water and land resource management faces unprecedented challenges. From the perspective of agricultural WEF-nexus relationships, achieving efficient allocation and sustainable utilization of agricultural water and land resources has become a focal issue in current agricultural development. This study contributes to the literature by introducing a constructing a bi-level multi-objective nonlinear programming model, overcoming the limitations of traditional single-level decision models. Our innovative model incorporates carbon emission management into the traditional WEF framework to form a more comprehensive WEFC nexus. Combining fuzzy goal programming and a non-dominated sorting elite strategy genetic algorithm, we explored the optimization of agricultural, water, and land resource allocation in the MRB. In addition to our theoretical contributions, our study provides practical findings summarized as follows:

(1) By achieving collaborative management of agricultural water and land resources through optimizing irrigation water and land resource allocation, our model results indicate that i) economic benefits maintained relatively stable fluctuations, ii) food production efficiency maintained a high level, and iii) unit food carbon emissions were controlled within a reasonable range, when considering different weights between upper-level and lower-level decision-makers.

(2) Our model reveals spatial variation in agricultural resource allocation across the MRB. Water resource allocation forms a "three-tier gradient" pattern: the Chengdu Plain area receives the maximum quota, followed by middle-stream regions (Leshan, Yibin), and minimum allocation in upstream mountainous areas. Carbon emission intensity also varies across areas: significant gaps exist between high-emission areas (e.g., Chengdu) and low-emission areas (e.g. Aba Prefecture), with CO indirect carbon emissions generally exceeding 80% (reaching 99.10% in Ya'an), while carbon sequestration effects show the Aba Prefecture far exceeding plain agricultural areas.

(3) Regarding crop planting structure, regions demonstrate distinct adaptive differentiation. In upstream areas, Chengdu focuses on rice as the dominant crop, while Aba Prefecture primarily grows wheat and corn. Middle and downstream regions show similar differentiation characteristics, with Yibin dominated by rice cultivation, while Ya'an maintains a relative balance between rice and corn cultivation. This spatial differentiation pattern fully embodies the principle of "adapting measures to local conditions" in resource allocation.

We propose three key policy recommendations to optimize resource management in the Minjiang River Basin. First, differentiated resource allocation strategies should be implemented to reflect regional heterogeneity. Our results indicate that intensive agriculture in high-resource zones (e.g., Chengdu), balanced strategies in medium-resource areas (e.g., Leshan, Yibin), and ecological protection with specialized agriculture in low-resource zones (e.g., Aba Prefecture) are optimal resource allocation strategies. Second, region-specific carbon management should set stricter controls in high-emission areas, optimize fertilizer use, and promote carbon sink projects in high-sequestration regions. Lastly, improving multi-level decision-making mechanisms is critical, with a coordinated governance structure where basin-level authorities oversee overall resource efficiency and environmental targets, while regional governments implement localized crop optimization strategies.

While our new WEFC framework provides new theoretical perspectives and methods for optimizing agricultural resource allocation, the model is limited in addressing overall climate change impacts and abstracts away from technological progress effects. Note that these limitations primarily concern the long-term dynamics of both the environment itself and the institutions that affect said environment. Nevertheless, the model and subsequent analysis provides a comprehensive and robust framework which focusses on the short/medium-run. Future research is required on deepening the assessment of the long-term impacts of on agricultural systems which enhances model applicability in complex decision environments whilst still promoting sustainable agricultural resource utilization and regional coordinated development.

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